



GIRIJANANDA CHOWDHURY UNIVERSITY
in association with
**NOVEL INTEGRATED ELECTRONICS LABS (NINE
LABS), IIT Guwahati**

Workshop Report on “VLSI Design Using Open Source Tools”

**Sponsored by
Ministry of Electronics and
Information Technology (MeitY))**

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LAB REPORT

Objective:

The objective of this experiment is to perform a complete RTL to GDSII physical design flow using the OpenROAD toolchain with the SkyWater130 nm (sky130) PDK. The design includes multi-metal routing stacks and reports key physical and timing parameters such as Setup Slack value Design area value, Core utilization value, TNS value, WNS value, max capacitance, max path delay, min capacitance, min path delay, Clock period used and Power consumption information in tabular format.

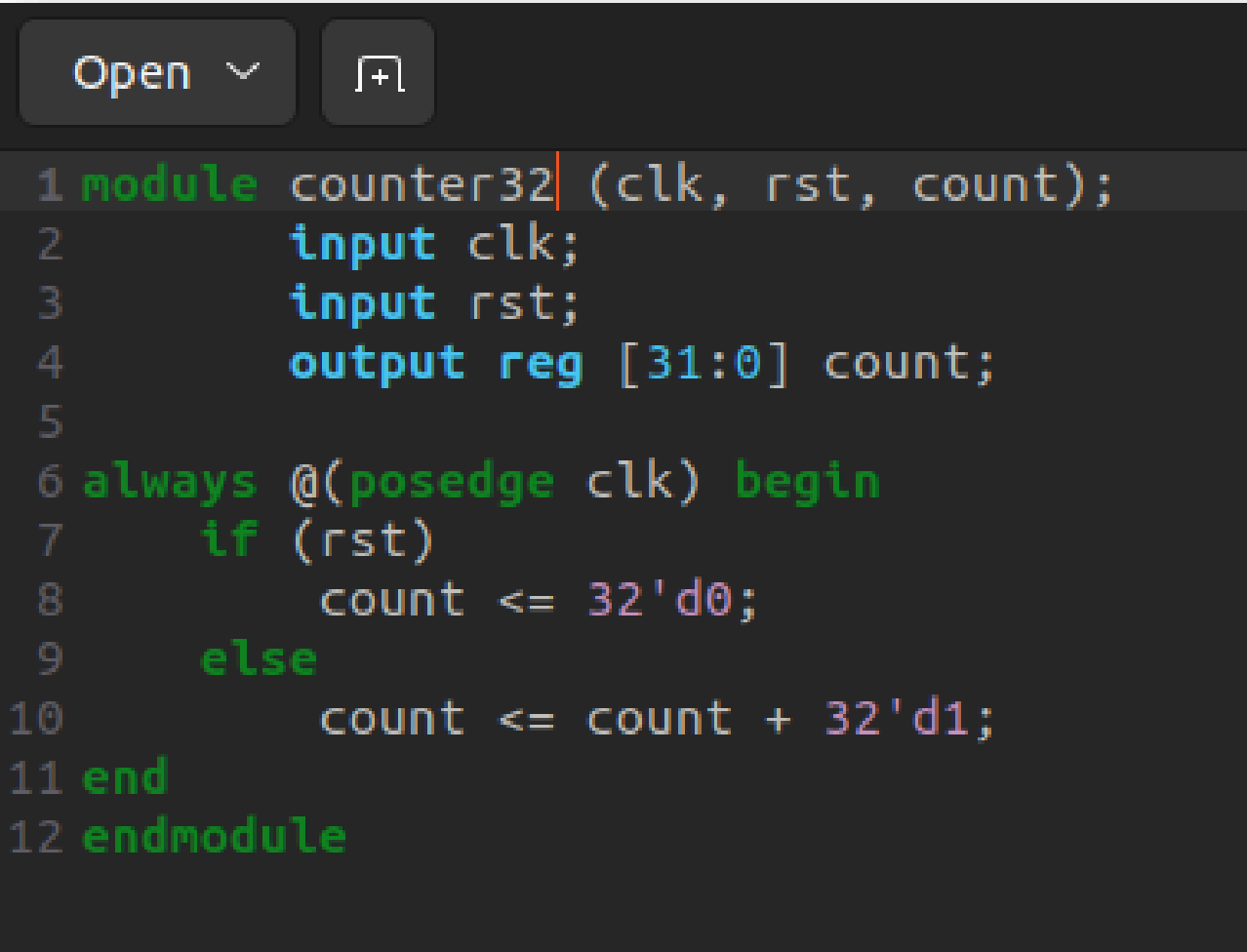
Tools and Technologies used:

Item	Description
EDA Tool	OpenROAD
Technology	SkyWater Sky130
OS	Ubuntu Linux

RTL Design description:

A 32-bit synchronous up-counter was chosen as the target design. The counter increments on every rising edge of the clock and resets asynchronously.

RTL Code: (counter32.v)



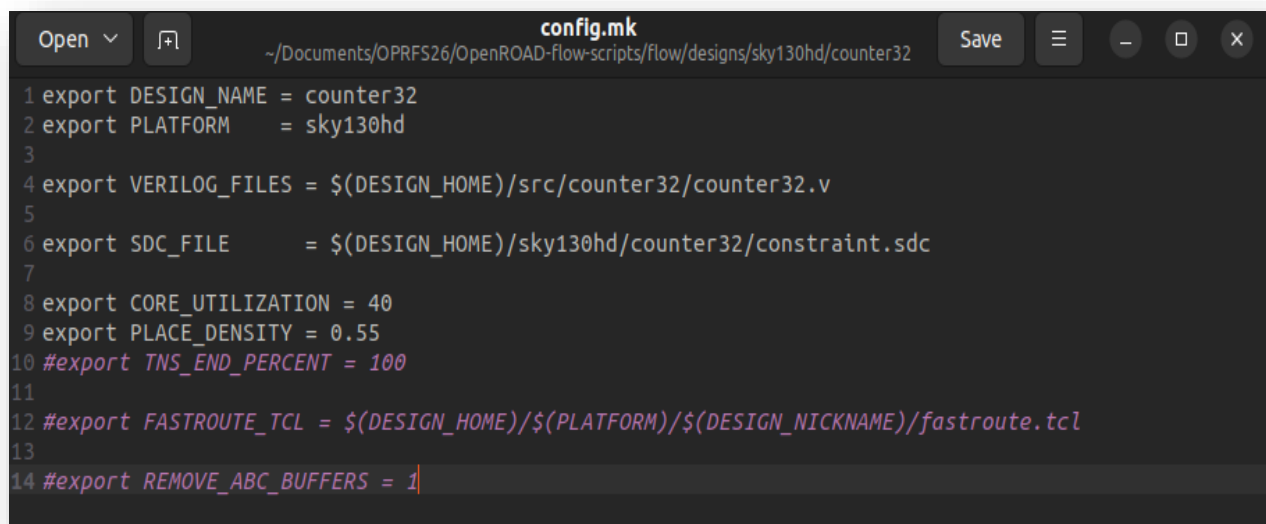
```
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```

```
1 module counter32 (clk, rst, count);
2     input clk;
3     input rst;
4     output reg [31:0] count;
5
6     always @(posedge clk) begin
7         if (rst)
8             count <= 32'd0;
9         else
10            count <= count + 32'd1;
11    end
12 endmodule
```

Configuration file (config.mk):

It defines the:

- Design name
- Platform (sky130hd)
- RTL path
- Core utilization and placement density

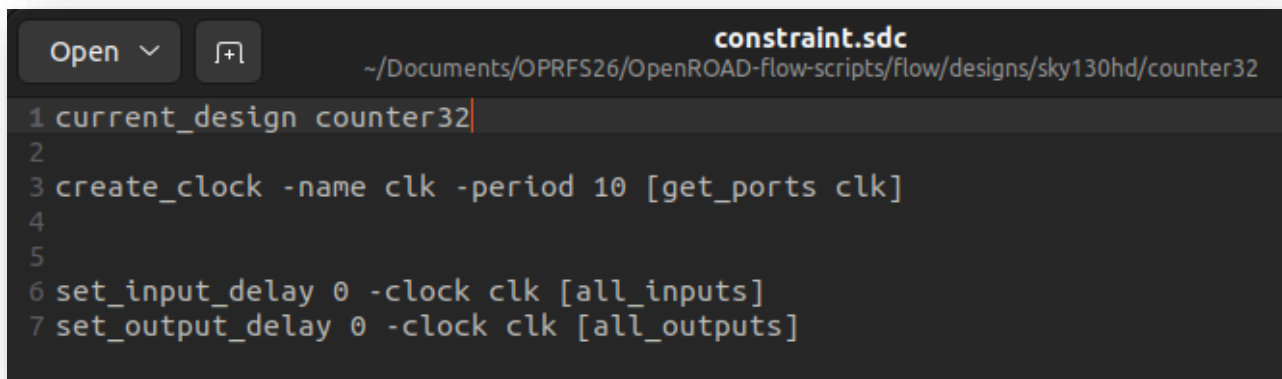
A screenshot of a text editor window titled "config.mk". The window has a dark background and shows a list of export statements for configuration. The path in the title bar is "~/Documents/OPRFS26/OpenROAD-flow-scripts/flow/designs/sky130hd/counter32". The code includes standard export statements for design name, platform, Verilog files, SDC file, core utilization, and placement density, along with commented-out options for TNS_END_PERCENT, FASTROUTE_TCL, and REMOVE_ABC_BUFFERS.

```
1 export DESIGN_NAME = counter32
2 export PLATFORM    = sky130hd
3
4 export VERILOG_FILES = $(DESIGN_HOME)/src/counter32/counter32.v
5
6 export SDC_FILE      = $(DESIGN_HOME)/sky130hd/counter32/constraint.sdc
7
8 export CORE_UTILIZATION = 40
9 export PLACE_DENSITY = 0.55
10 #export TNS_END_PERCENT = 100
11
12 #export FASTROUTE_TCL = $(DESIGN_HOME)/$(PLATFORM)/$(DESIGN_NICKNAME)/fastroute.tcl
13
14 #export REMOVE_ABC_BUFFERS = 1
```

Timing Constraint (constraint.sdc):

It defines the:

- Clock period = 10 ns
- Input and output delays

A screenshot of a text editor window titled "constraint.sdc". The window has a dark background and a light-colored border. The file path is shown as "~/Documents/OPRFS26/OpenROAD-flow-scripts/flow/designs/sky130hd/counter32". The editor contains the following text:

```
1 current_design counter32|
2
3 create_clock -name clk -period 10 [get_ports clk]
4
5
6 set_input_delay 0 -clock clk [all_inputs]
7 set_output_delay 0 -clock clk [all_outputs]
```

Procedure:

Step 1: RTL Preparation

- Created Verilog design of 32-bit counter
 - Verified syntax and functionality
-

Step 2: Configuration Setup

- Edited config.mk to set:
 - DESIGN_NAME = counter32
 - PLATFORM = sky130hd
 - CORE_UTILIZATION = 40
 - PLACE_DENSITY = 0.55
-

Step 3: Timing Constraint Definition

- Created Constraint.sdc file.
- Defined Clock period as 10ns.
- Applied input/output delays

Step 4: Running OpenROAD Flow

The following RTL-to-GDS flow was executed:

1. Read RTL Verilog
2. Apply timing constraints
3. Floorplanning
4. Placement
5. Clock Tree Synthesis (CTS)
6. Global routing
7. Detailed routing (multi-metal stack)
8. GDSII generation

Parameters reported:

Setup Slack Value, TNS and WNS value:

Path Group: clk
Path Type: max

Cap	Delay	Time	Description

	0.00	0.00	clock clk (rise edge)
	0.32	0.32	clock network delay (propagated)
	0.00	0.32	^ count[4]\$_SDFF_PP0_/CLK (sky130_fd_sc_hd_dfxtp_1)
0.02	0.44	0.76	^ count[4]\$_SDFF_PP0_/Q (sky130_fd_sc_hd_dfxtp_1)
0.03	0.31	1.07	v _138_/Y (sky130_fd_sc_hd_nand3_1)
0.05	0.84	1.92	v _168_/X (sky130_fd_sc_hd_or4_1)
0.00	0.58	2.49	v _227_/X (sky130_fd_sc_hd_or4_1)
0.00	0.15	2.64	^ _228_/Y (sky130_fd_sc_hd_a2loi_1)
	0.00	2.64	^ count[28]\$_SDFF_PP0_/D (sky130_fd_sc_hd_dfxtp_1)
		2.64	data arrival time

	10.00	10.00	clock clk (rise edge)
	0.32	10.32	clock network delay (propagated)
	0.00	10.32	clock reconvergence pessimism
		10.32	^ count[28]\$_SDFF_PP0_/CLK (sky130_fd_sc_hd_dfxtp_1)
	-0.06	10.26	library setup time
		10.26	data required time

		10.26	data required time
		-2.64	data arrival time

		7.62	slack (MET)

```
6_finish.rpt [Read-Only]
~/Documents/OPRFS26/OpenROAD-flow-scripts/flow/reports/sky130hd/counter32/base
Open  Save  -  □  ×

1
2 =====
3 finish report_tns
4 -----
5 tns max 0.00
6
7 =====
8 finish report_wns
9 -----
10 wns max 0.00
11
12 =====
13 finish report_worst_slack
14 -----
15 worst slack max 7.62
16
```

Design area and Core Utilization Value:

Scripting

```
gui::select_chart "Endpoint Slack"
gui::update_timing_report
>>> report_design_area
Design area 1800 um^2 53% utilization.
```

Maximum path delay:

Endpoint: count[28]\$_SDFF_PP0_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Cap	Slew	Delay	Time	Description

		0.000	0.000	clock clk (rise edge)
		0.320	0.320	clock network delay (propagated)
	0.090	0.000	0.320	^ count[4]\$_SDFF_PP0_/CLK (sky130_fd_sc_hd_dfxt1)
0.021	0.204	0.436	0.756	^ count[4]\$_SDFF_PP0_/Q (sky130_fd_sc_hd_dfxt1)
0.031	0.317	0.315	1.071	v _138_/Y (sky130_fd_sc_hd_nand3_1)
0.046	0.290	0.845	1.915	v _168_/X (sky130_fd_sc_hd_or4_1)
0.003	0.083	0.576	2.492	v _227_/X (sky130_fd_sc_hd_or4_1)
0.003	0.107	0.150	2.641	v _228_/Y (sky130_fd_sc_hd_a2loi_1)
	0.107	0.000	2.641	^ count[28]\$_SDFF_PP0_/D (sky130_fd_sc_hd_dfxt1)
			2.641	data arrival time

		10.000	10.000	clock clk (rise edge)
		0.319	10.319	clock network delay (propagated)
		0.000	10.319	clock reconvergence pessimism
			10.319	^ count[28]\$_SDFF_PP0_/CLK (sky130_fd_sc_hd_dfxt1)
		-0.062	10.257	library setup time
			10.257	data required time

			10.257	data required time
			-2.641	data arrival time

			7.616	slack (MET)

Minimum path delay:

```
sta::find_timing
sta::find_requireds
gui::select_chart "Endpoint Slack"
gui::update_timing_report
[WARNING STA-0168] unknown field delay.
Startpoint: rst (input port clocked by clk)
Endpoint: count[2]$_SDFF_PP0_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
```

Cap	Slew	Delay	Time	Description
		0.000	0.000	clock clk (rise edge)
		0.000	0.000	clock network delay (propagated)
		0.000	0.000	v input external delay
0.002	0.000	0.000	0.000	v rst (in)
0.012	0.111	0.460	0.460	v input1/X (sky130_fd_sc_hd__clkdlybuf4s50_1)
0.002	0.066	0.113	0.574	v _234_/Y (sky130_fd_sc_hd__nor2_1)
	0.066	0.000	0.574	^ count[2]\$_SDFF_PP0_/D (sky130_fd_sc_hd__dfxtp_1)
			0.574	data arrival time
		0.000	0.000	clock clk (rise edge)
		0.320	0.320	clock network delay (propagated)
		0.000	0.320	clock reconvergence pessimism
			0.320	^ count[2]\$_SDFF_PP0_/CLK (sky130_fd_sc_hd__dfxtp_1)
		-0.030	0.291	library hold time
			0.291	data required time
			0.291	data required time
			-0.574	data arrival time
			0.283	slack (MET)

Maximum capacitance:

```
gui::update_timing_report
>>> report_check_types -min_capacitance
>>> report_check_types -max_capacitance
max capacitance
```

Pin	Limit	Cap	Slack
158/Y	0.04	0.01	0.03 (MET)

Minimum capacitance: It is not applicable, as the earlier TCL command `report_checks -min capacitance` returned no results.

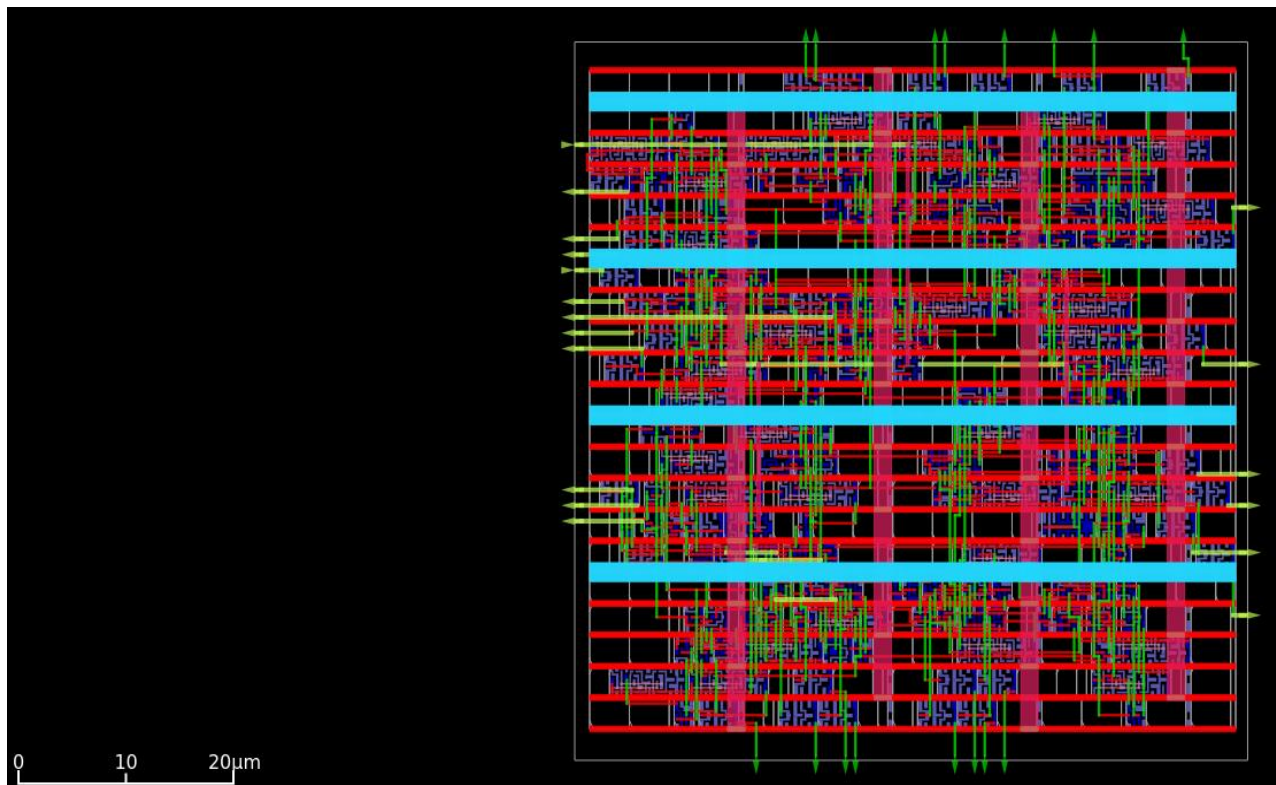
Clock period used & Power consumption information in tabular format:

```
>>> get_property [all_clocks] period  
10.000000
```

```
>>> report_power
```

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)	
Sequential	1.28e-04	1.35e-06	2.58e-10	1.30e-04	61.2%
Combinational	3.17e-06	2.99e-06	2.57e-10	6.16e-06	2.9%
Clock	3.47e-05	4.15e-05	3.26e-11	7.61e-05	35.9%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	1.66e-04	4.58e-05	5.47e-10	2.12e-04	100.0%
	78.4%	21.6%	0.0%		

Image of GDS Layout:



Conclusion:

In this experiment, a complete RTL to GDSII physical implementation was successfully carried out using the OpenROAD toolchain and Sky130 PDK. All required timing, power, and physical parameters were met, and the design was successfully routed using multiple metal layers. This demonstrates the feasibility of an open-source ASIC design flow from RTL to manufacturable layout.

Execution of RTL-to-GDSII Flow Using OpenROAD:

```
bishhnu@bishhnu-VirtualBox:~$ cd Documents/OPRFS26/OpenROAD-flow-scripts/flow/designs/src/
bishhnu@bishhnu-VirtualBox:~/Documents/OPRFS26/OpenROAD-flow-scripts/flow/designs/src$ cd counter32/
bishhnu@bishhnu-VirtualBox:~/Documents/OPRFS26/OpenROAD-flow-scripts/flow/designs/src/counter32$ gedit counter32.v
bishhnu@bishhnu-VirtualBox:~/Documents/OPRFS26/OpenROAD-flow-scripts/flow/designs/src/counter32$ cd ../../
bishhnu@bishhnu-VirtualBox:~/Documents/OPRFS26/OpenROAD-flow-scripts/flow/designs$ cd sky130hd/counter32/
bishhnu@bishhnu-VirtualBox:~/Documents/OPRFS26/OpenROAD-flow-scripts/flow/designs/sky130hd/counter32$ gedit config.mk
bishhnu@bishhnu-VirtualBox:~/Documents/OPRFS26/OpenROAD-flow-scripts/flow/designs/sky130hd/counter32$ gedit constraint.sdc
bishhnu@bishhnu-VirtualBox:~/Documents/OPRFS26/OpenROAD-flow-scripts/flow/designs/sky130hd/counter32$ cd ../../
bishhnu@bishhnu-VirtualBox:~/Documents/OPRFS26/OpenROAD-flow-scripts/flow$ export OR_IMAGE=docker.io/openroad/orfs:v3.0-4602-gfcb70dad
bishhnu@bishhnu-VirtualBox:~/Documents/OPRFS26/OpenROAD-flow-scripts/flow$ sudo ./util/docker_shell make DESIGN_CONFIG=./designs/sky130hd/counter32/config.mk
[sudo] password for bishhnu:
+++ dirname ./util/docker_shell
++ cd ./util
++ pwd
+ DIR=/home/bishhnu/Documents/OPRFS26/OpenROAD-flow-scripts/flow/util
+ get_container_engine
++ docker --version
+ local 'DOCKER_VERSION_STRING=Docker version 28.2.2, build 28.2.2-0ubuntu1~22.04.1'
+ [[ Docker version 28.2.2, build 28.2.2-0ubuntu1~22.04.1 == *\D\o\c\k\e\r* ]]
+ container_engine=docker
++ pwd
+ WORKSPACE=/home/bishhnu/Documents/OPRFS26/OpenROAD-flow-scripts/flow
+ YOSYS_EXE=/OpenROAD-flow-scripts/tools/install/yosys/bin/yosys
+ OPENROAD_EXE=/OpenROAD-flow-scripts/tools/install/OpenROAD/bin/openroad
+ KLAYOUT_CMD=/usr/bin/klayout
+ XSOCK=/tmp/.X11-unix
+ XAUTH=/tmp/.docker.xauth
+ xauth nlist :0
+ sed -e 's/^.../ffff/'
+ xauth -f /tmp/.docker.xauth nmerge -
+ ARGUMENTS='make DESIGN_CONFIG=./designs/sky130hd/counter32/config.mk'
+ test -t 0
DOCKER_INTERACTIVE=+1
```

```
bishhnu@bishhnu-VirtualBox:~/Documents/OPRFS26/OpenROAD-flow-scripts/flow$ sudo ./util/docker_shell make DESIGN_CONFIG=./designs/sky130hd/counter32/config.mk gui_final
+++ dirname ./util/docker_shell
++ cd ./util
++ pwd
+ DIR=/home/bishhnu/Documents/OPRFS26/OpenROAD-flow-scripts/flow/util
+ get_container_engine
++ docker --version
+ local 'DOCKER_VERSION_STRING=Docker version 28.2.2, build 28.2.2-0ubuntu1~22.04.1'
+ [[ Docker version 28.2.2, build 28.2.2-0ubuntu1~22.04.1 == *\D\o\c\k\e\r* ]]
+ container_engine=docker
++ pwd
+ WORKSPACE=/home/bishhnu/Documents/OPRFS26/OpenROAD-flow-scripts/flow
+ YOSYS_EXE=/OpenROAD-flow-scripts/tools/install/yosys/bin/yosys
+ OPENROAD_EXE=/OpenROAD-flow-scripts/tools/install/OpenROAD/bin/openroad
+ KLAYOUT_CMD=/usr/bin/klayout
+ XSOCK=/tmp/.X11-unix
+ XAUTH=/tmp/.docker.xauth
+ xauth nlist :0
+ sed -e 's/^.../ffff/'
+ xauth -f /tmp/.docker.xauth nmerge -
+ ARGUMENTS='make DESIGN_CONFIG=./designs/sky130hd/counter32/config.mk gui_final'
+ test -t 0
+ DOCKER_INTERACTIVE=-t
+ [[ docker == \p\o\d\m\a\n ]]
++ id -u root
++ id -g root
++ user args='-u 0:0'
+ docker run -u 0:0 -e LIBGL_ALWAYS_SOFTWARE=1 -e QT_X11_NO_MITSHM=1 -e XDG_RUNTIME_DIR=/tmp/xdg-run -e DISPLAY=:0 -e QT_XKB_CONFIG_ROOT=/usr/share/X11/xkb -v /tmp/.X11-unix:/tmp/.X11-unix -v /tmp/.docker.xauth:/tmp/.docker.xauth -e XAUTHORITY=/tmp/.docker.xauth -e FLOW_HOME=/OpenROAD-flow-scripts/flow/ -e YOSYS_EXE=/OpenROAD-flow-scripts/tools/install/yosys/bin/yosys -e OPENROAD_EXE=/OpenROAD-flow-scripts/tools/install/OpenROAD/bin/openroad -e KLAYOUT_CMD=/usr/bin/klayout -v /home/bishhnu/Documents/OPRFS26/OpenROAD-flow-scripts/flow/designs/sky130hd/counter32
```



```

bishhnu@bishhnu-VirtualBox: ~/Documents/OPRFS26/OpenROAD-flow-scripts/flow/designs/sky130hd/counter32
-rd out_file=./results/sky130hd/counter32/base/6_1_merged.gds \
-rd tech_file=./objects/sky130hd/counter32/base/klayout.lyt \
-rd layer_map= \
-r /OpenROAD-flow-scripts/flow/util/def2stream.py) 2>&1 | tee /OpenROAD-flow-scripts/flow/logs/sky130hd/counter32/base/6_1
merge.log
KLayout 0.30.3
[INFO] Reporting cells prior to loading DEF ...
/OpenROAD-flow-scripts/flow/platforms/sky130hd/gds/sky130_fd_sc_hd.gds
[INFO] All LEF cells have matching GDS/OAS cells
[INFO] No orphan cells in the final layout
Elapsed time: 0:02.05[h:min:sec. CPU time: user 1.81 sys 0.19 (98%). Peak memory: 436680KB.
cp results/sky130hd/counter32/base/6_1_merged.gds results/sky130hd/counter32/base/6_final.gds
./logs/sky130hd/counter32/base
Log
Elapsed/s Peak Memory/MB sha1sum .odb [0:20]
1_1_yosys_canonicalize 0 52 1608d6594f17eb2f7133
1_2_yosys 1 55 63b1c280fcb5bf29507c
1_synth 0 115 f5e4eba59d3cb26f6f3a
2_1_floorplan 0 135 2b5134e2548a92e47257
2_2_floorplan_macro 0 113 2b5134e2548a92e47257
2_3_floorplan_tapcell 0 113 72a68f6df6076516348c
2_4_floorplan_pdn 0 115 f0756328181a47144a66
3_1_place_gp_skip_io 0 113 cbb52c83d4ad1493bc80
3_2_place_iop 0 113 76bd235bacba4ca58fb9
3_3_place_gp 1 219 b042cfd5540cb2ede4e2
3_4_place_resized 0 134 b042cfd5540cb2ede4e2
3_5_place_dp 0 117 84ce9a0ca9b3f8bb023d
4_1_cts 4 159 f879f484cb89c0ef9ac4
5_1_grt 3 227 b7860b41dbe01947475d
5_2_route 12 632 512e546fb69a4f90de86
5_3_fillcell 0 115 7cac76940e4f3c8d4d8c
6_1_fill 0 113 7cac76940e4f3c8d4d8c
6_1_merge 2 426 N/A
6_report 2 231 N/A
Total 25 632

```

```

bishhnu@bishhnu-VirtualBox:~/Documents/OPRFS26/OpenROAD-flow-scripts/flow$ sudo ./util/docker_shell make DESIGN_CONFIG=./designs/s
ky130hd/counter32/config.mk gui_final
+++ dirname ./util/docker_shell
++ cd ./util
++ pwd
+ DIR=/home/bishhnu/Documents/OPRFS26/OpenROAD-flow-scripts/flow/util
+ get_container_engine
++ docker --version
+ local 'DOCKER_VERSION_STRING=Docker version 28.2.2, build 28.2.2-0ubuntu1-22.04.1'
+ [[ Docker version 28.2.2, build 28.2.2-0ubuntu1-22.04.1 == *\D\o\c\k\e\r* ]]
+ container_engine=docker
++ pwd
+ WORKSPACE=/home/bishhnu/Documents/OPRFS26/OpenROAD-flow-scripts/flow
+ YOSYS_EXE=/OpenROAD-flow-scripts/tools/install/yosys/bin/yosys
+ OPENROAD_EXE=/OpenROAD-flow-scripts/tools/install/OpenROAD/bin/openroad
+ KLAYOUT_CMD=/usr/bin/klayout
+ XSOCK=/tmp/.X11-unix
+ XAUTH=/tmp/.docker.xauth
+ xauth nlist :0
+ sed -e 's/^...../ffff/'
+ xauth -f /tmp/.docker.xauth nmerge -
+ ARGUMENTS='make DESIGN_CONFIG=./designs/sky130hd/counter32/config.mk gui_final'
+ test -t 0
+ DOCKER_INTERACTIVE=-ti
+ [[ docker == \p\o\d\m\ \n ]]
++ id -u root
++ id -g root
+ user_args='-u 0:0'
+ docker run -u 0:0 -e LIBGL_ALWAYS_SOFTWARE=1 -e QT_X11_NO_MITSHM=1 -e XDG_RUNTIME_DIR=/tmp/xdg-run -e DISPLAY=:0 -e QT_XKB_CONFI
G_ROOT=/usr/share/X11/xkb -v /tmp/.X11-unix:/tmp/.X11-unix -v /tmp/.docker.xauth:/tmp/.docker.xauth -e XAUTHORITY=/tmp/.docker.xau
th -e FLOW_HOME=/OpenROAD-flow-scripts/flow/ -e YOSYS_EXE=/OpenROAD-flow-scripts/tools/install/yosys/bin/yosys -e OPENROAD_EXE=/Op
enROAD-flow-scripts/tools/install/OpenROAD/bin/openroad -e KLAYOUT_CMD=/usr/bin/klayout -v /home/bishhnu/Documents/OPRFS26/OpenROA
D-flow-scripts/flow:/OpenROAD-flow-scripts/flow:Z --network host -ti openroad/orfs:latest bash -c 'set -ex
mkdir /tmp/xdg-run

```

